

Imperial-Cambridge Math Contest 2024 Rd 2 — P1/5

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- (a) Prove that there exist distinct positive integers $a_1, a_2, \dots, a_{2024}$ such that for each $i \in \{1, 2, \dots, 2024\}$, a_i divides $a_1 a_2 \cdots a_{i-1} a_{i+1} \cdots a_{2024} + 1$.
- (b) Prove that there exist distinct positive integers $b_1, b_2, \dots, b_{2024}$ such that for each $i \in \{1, 2, \dots, 2024\}$, b_i divides $b_1 b_2 \cdots b_{i-1} b_{i+1} \cdots b_{2024} + 2024$.

Solution

(a): Let $a_1 = 3$, then define a_n like so $a_i = (a_1 a_2 \cdots a_{i-1}) + f(i)$, $i \geq 2$ where $f : \mathbb{N} \rightarrow \{-1, 1\}$ is 1 if i is even, and -1 if i is odd.

Now if i is odd, then there are $2024 - i$ numbers in the set $\{a_{i+1}, \dots, a_{2024}\}$. These numbers alternate between ± 1 modulo a_i and since there are an odd number of terms the product of elements in that set is -1 modulo a_i . Thus,

$$a_1 a_2 \cdots a_{i-1} a_{i+1} \cdots a_{2024} + 1 \equiv (a_i + 1)(-1) + 1 \equiv 0 \pmod{a_i}.$$

Similarly if i is even, then the product of the elements in the set $\{a_{i+1}, \dots, a_{2024}\}$ is 1 modulo a_i because there is an even number of elements in that set, and so

$$a_1 a_2 \cdots a_{i-1} a_{i+1} \cdots a_{2024} + 1 \equiv (a_i - 1)(1) + 1 \equiv 0 \pmod{a_i}.$$

To finish we just need to prove that all the a_i 's are distinct. We notice that

$$\begin{aligned} a_i &\geq (a_1 a_2 \cdots a_{i-1}) - 1 \\ &= (a_{i-1} \pm 1) a_{i-1} - 1 \\ &\geq (a_{i-1} - 1) a_{i-1} - 1 \\ &= (a_{i-1})^2 - a_{i-1} - 1 \end{aligned}$$

But since $x^2 - x - 1 > x$ holds for $x > 1 + \sqrt{2}$ and $a_1 = 3 > 1 + \sqrt{2}$, we have that $a_i \geq (a_{i-1})^2 - a_{i-1} - 1 > a_{i-1}$ and so (a_n) is strictly increasing. This completes the proof. $\square$

(b): We use a similar approach. Let $b_1 = 4048$, then define b_n like so $b_i = (b_1 b_2 \cdots b_{i-1}) + f(i)$, $2 \leq i \leq 2022$, where $f(i)$ has the same definition as in part (a). Set $b_{2023} = 1$ and $b_{2024} = 2024$.

If $i = 2023$ then we have nothing to prove since every positive integer is divisible by 1. If $i = 2024$ then $a_1 \cdots a_{2023} + 2024 \equiv 0 \pmod{b_{2024}}$ because $b_1 = 4048 = 2 \cdot 2024$. Now in what proceeds assume $i \leq 2022$.

If i is odd then there are $2022 - i$ numbers in the set $\{b_{i+1}, \dots, b_{2022}\}$. These numbers alternate between ± 1 modulo b_i and since there is an odd number of terms, the product of elements in that set is $-1 \pmod{b_i}$. Thus we write

$$\begin{aligned}
& b_1 b_2 \cdots b_{i-1} b_{i+1} \cdots b_{2024} + 2024 \\
& \equiv (b_i + 1)(-1) \cdot b_{2023} \cdot b_{2024} + 2024 \\
& \equiv -2024 + 2024 \equiv 0 \pmod{b_i}
\end{aligned}$$

Now if i is even, then the product of the elements in the set $\{b_{i+1}, \dots, b_{2022}\}$ is 1 mod b_i since there is an even number of elements in that set. So we put

$$\begin{aligned}
& b_1 b_2 \cdots b_{i-1} b_{i+1} \cdots b_{2024} + 2024 \\
& \equiv (b_i - 1)(1) \cdot b_{2023} \cdot b_{2024} + 2024 \\
& \equiv -2024 + 2024 \equiv 0 \pmod{b_i}
\end{aligned}$$

Now due to the argument in (a) we know that the numbers $a_1, \dots, a_{2022}$ are all pairwise distinct and strictly greater than 4047 and since 1 and 2024 are distinct and both strictly less than 4047, the entire sequence is pairwise distinct as desired. This completes the proof. ■